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CSIT127
Networks and Communication
PROJECT REPORT

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Table of Contents

Introduction	4
Objectives	4
Business Requirement Analysis	5
Organization Overview:.....	5
Departmental requirements:	5
Communication requirements:.....	5
Internet and wireless requirements:.....	6
Security requirements	6
Scalability and availability requirements:.....	6
Network Design.....	7
Network Overview	7
Network Topology.....	7
Logical diagram:	7
IP Addressing Plan:.....	8
Screenshot of Main Switch VLAN Configuration (show vlan brief):.....	8
Screenshot of Router Interface Configuration (show ip interface brief):.....	9
Security Design.....	10
VLAN Segmentation:	10
Access Control Lists (ACLs):.....	10
Firewall Security:.....	10
Router Security:	11
Wireless Security:.....	11
Screenshot of Staff Access point Configuration:.....	12
Screenshot of Guest Access point configuration:	13
Implementation & Testing	15
Router-on-a-Stick Implementation:	15
Screenshots of Router Sub interfaces Configuration:.....	15
VLAN and Trunking Configuration:	16
Access Ports and Physical Topology:.....	17
NAT Configuration:	17
Screenshot - NAT Translation Table:.....	17
Network Design.....	18
Connectivity Testing.....	18
Screenshots– Successful Pings to Internet (8.8.8.8):	19
Guest wireless device:.....	19
Security Testing.....	20
Screenshots of HR pc pinging Finance failing (ACL block):	20

Screenshot- Guest devices cannot ping internal devices(ACL BLOCK):	22
Screenshots – Guest wireless devices cannot ping Staff wireless devices(& vice versa):	22
Screenshot – Guest to Guest Successful ping:	23
Screenshot - Staff to Staff Successful ping:.....	24
Screenshot – ACL Configuration	25
Successful pings:	26
Failed pings-ACL Blocking Test:.....	27
Screenshot of Server DHCP:.....	28
Screenshots- Router Security Configuration (show running-config):	29
Screenshots for VLAN segmentation in each department switch:	31
Screenshot of the Routing Table:.....	34
Discussion & Evaluation	35
Conclusion.....	36
Reference List:.....	37
Work Distribution:	38

Introduction

This project is on the design and implementation of a secure enterprise-level computer network for a medium-sized organization, especially an IT services firm. Today's organizations rely largely on efficient and secure networks to enable everyday operations, internal communication, and access to digital resources. As a company grows, its network infrastructure must be scalable, dependable, and capable of servicing numerous departments with diverse access needs.

The suggested network design meets these requirements by using industry-standard networking technologies such as routers, Layer 2 and Layer 3 switches, Virtual Local Area Networks (VLANs), wireless access points, and centralized servers.

Security is a major priority in this architecture, with technologies like access control lists (ACLs) and network segmentation used to safeguard critical data and prevent unauthorized access.

The network will be logically built and simulated with Cisco Packet Tracer, which will enable the installation and testing of IP addressing, inter-VLAN routing, wireless connection, and security rules. This project demonstrates how a well-structured enterprise network can improve performance, increase security, and allow for future development.

Objectives

The primary aims of this initiative are as follows:

1. To assess the networking needs of a medium-sized IT services organization with many divisions.
2. Design a logical business network topology that enables safe and efficient internal communication.
3. To use VLANs to separate departmental traffic and increase network security and performance.
4. To set up routing methods that allow communication between distinct VLANs.
5. To create and implement a suitable IP addressing scheme utilizing DHCP for effective network administration.
6. To implement wireless networking for staff and visitor users with restricted access.
7. To prevent unauthorized access, use basic network security techniques such as access control lists (ACLs).
8. To ensure that the network design is scalable, dependable, and adaptable to future organizational expansion.
9. To simulate and evaluate the whole network design with Cisco Packet Tracer.

Business Requirement Analysis

Organization Overview:

The company is a medium-sized IT service provider which operates from a central office. Its day to day operations rely heavily on internet connectivity, internal information system, centralized servers, and digital communication tools. To meet current requirements of operation and support growth in future the company must have secure, stable and scalable network infrastructure.

A strong, well thought out enterprise network is the backbone of any business, keeping operations running smoothly, safeguarding critical data, and making sure teams can communicate effortlessly. Kurose and Rose (2011) stress that for a network to truly work, it needs to be reliable, flexible, and secure-so the business never skips a beat. Oppenheimer (2011) adds a network shouldn't just handle today's demands. It has to be smart enough to grow with the company and support its long term vision.

Departmental requirements:

In any organization, different departments have different roles, responsibilities, and data needs. Think of departments like HR, Finance, IT and Sales Management- each has its own “personality” and requires the network to work in ways that match what they do.

The HR and Finance departments manage confidential employee records and financial data. Therefore, their network access must have restrictions and be closely observed. Implementing access control mechanism and traffic segmentation helps in assisting unauthorized access. (Stallings, 2020).

The Sales departments need continuous access to internal systems and internet services for customer relationship management, communications, and business dealings. Reliable and consistent network execution is necessary to support these activities (Kurose and Ross, 2021)

Communication requirements:

Good communication within the company is key for teamwork, sharing files, and using the central system efficiently. The network should make sure this works smoothly.

Firm network design principles must note the importance of backup, traffic management, and performance improvement to maintain service availability (Oppenheimer, 2011). In addition to this proper mechanism and switching mechanism contribute to efficient data delivery across internal networks (Kurose and Ross, 2021).

Setting up firewalls and using simple rules helps to protect both wired and wireless networks from threats coming from inside and outside. Firewalls are important for network safety.

Internet and wireless requirements:

Internet access must be managed according to the needs of departments. HR and Finance should have restricted internet exposure to reduce security risks, while IT and sales require broader access for operational efficiency.

Wireless connections must be available for both employees and visitors. Staff wireless networks should merge securely with internal systems and enforce authentication mechanisms such as WPA2 or enterprise level authentication protocols (Gast, 2021).

Guest wireless access must be logically separated from the internal network and limited strictly to internet access. Network segmentation and firewall policies should prevent users from connecting to internal organizational access (Scarfone and Hoffman, 2009).

Proper firewall setup and access control rules are critical to protecting both wired and wireless networks from internal and external threats (Scarfone and Hoffman, 2009).

Security requirements

Since the team handles personal business and employee data, strong safety controls must be applied. The network plan should include:

- Departmental traffic separation

- Firewall implementation

- Secure wireless authentication

- Network security

Network safety rules confidentiality, integrity, and availability

(CIA triad) as fundamental protection objectives (Stallings, 2020). Firewalls and reach control lists (ACLs) help regulate traffic flow and avoid unauthorized access (Scarfone and Hoffman, 2009).

Scalability and availability requirements:

The network needs to be able to handle growth. This means it has to work with employees and departments. It also has to be able to handle devices and services. The network should be able to do all of this without needing an update. We do not want to have to make changes to the network every time we add something new. The network must support growth, including additional employees and extra devices and added services without requiring a major redesign. Scalable network plans allow teams to grow effectively while maintaining work (Oppenheimer, 2011).

It is really important to have availability so that the system does not go down. This means we need to have links, backup systems and a good network design. All these things help to make sure the service is always available. For example Kurose and Ross said this in 2021. High availability is what we need to make sure everything runs smoothly (Kurose and Ross, 2021).

A scalable and highly available system ensures long-term running stability and supports business growth.

Network Design

Network Overview

The network design simulates a secure network environment for a medium-sized organization. The organization has different departments, i.e., Human Resources, Finance, Information Technology, Sales, Management, and Guest.

For the network to communicate efficiently and securely, the network is segmented using VLANs and router-on-a-stick inter-vlan routing. Each department is placed in a separate VLAN to reduce broadcast traffic and enforce network security policies.

The network has the following devices:

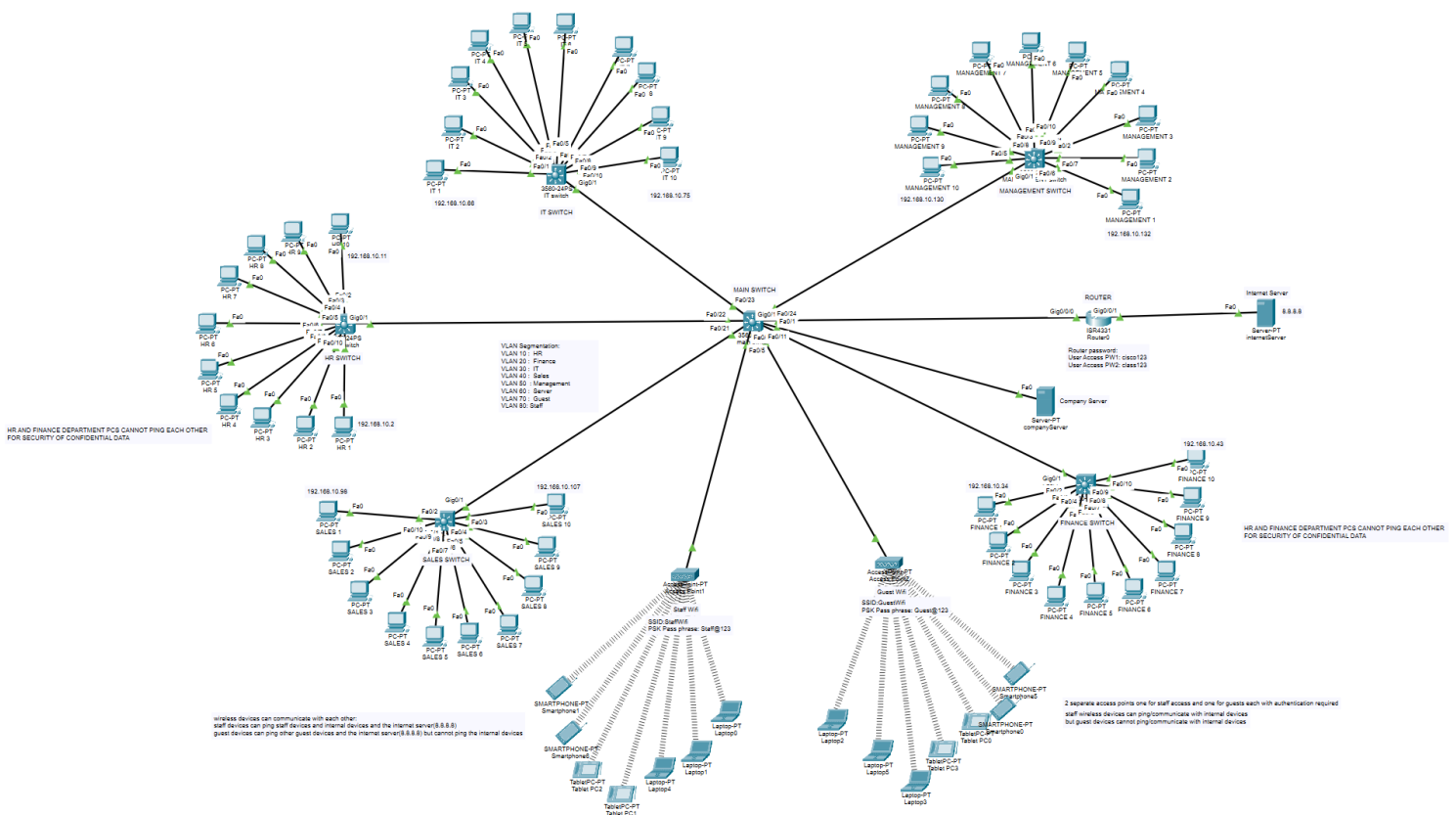
- A router to provide routing and internet access
- A main switch to provide VLAN services that connects to a switch for each department
- PCs to represent the different departments
- A main server to provide services to the organization
- A simulated internet server.
- wireless access points to provide wireless connectivity and wireless staff & guest devices.

Network Topology

The network topology is a star-shaped hierarchical network. The router is connected to the main switch using a trunk link. The router performs inter-vlan routing using sub interfaces. Each VLAN is used to represent a different department in the organization, thus providing a logical separation of the network while using the same switch.

The router is also used to provide Network Address Translation (NAT) services, allowing the organization's users to access the simulated internet server.

Logical diagram:



IP Addressing Plan:

VLAN	Department	Network Address	Subnet Mask	Default Gateway	Usable Host Range	Broadcast
10	HR	192.168.10.0	255.255.255.224	192.168.10.1	192.168.10.1 – 192.168.10.30	192.168.10.31
20	Finance	192.168.10.32	255.255.255.224	192.168.10.33	192.168.10.33 – 192.168.10.62	192.168.10.63
30	IT	192.168.10.64	255.255.255.224	192.168.10.65	192.168.10.65 – 192.168.10.94	192.168.10.95
40	Sales	192.168.10.96	255.255.255.224	192.168.10.97	192.168.10.97 – 192.168.10.126	192.168.10.127
50	Management	192.168.10.128	255.255.255.224	192.168.10.129	192.168.10.129– 192.168.10.158	192.168.10.159
60	Servers	192.168.10.160	255.255.255.224	192.168.10.161	192.168.10.161– 192.168.10.190	192.168.10.191
70	Guest	192.168.10.192	255.255.255.224	192.168.10.193	192.168.10.193– 192.168.10.222	192.168.10.223
80	Staff	192.168.10.224	255.255.255.224	192.168.10.225	192.168.10.225– 192.168.10.254	192.168.10.255

Screenshot of Main Switch VLAN Configuration (show vlan brief):

The screenshot shows a network switch CLI window titled "main switch". The "CLI" tab is selected, displaying the "IOS Command Line Interface". The output shows several status messages for interfaces FastEthernet0/11, FastEthernet0/1, FastEthernet0/5, and FastEthernet0/24, indicating link changes and protocol status updates. Below this, the command "Switch>show vlan brief" is executed, resulting in a table of VLAN configurations.

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Gig0/2
10	HR	active	
20	Finance	active	
30	IT	active	Fa0/10
40	sales	active	
50	Management	active	
60	servers	active	Fa0/1
70	Guest	active	Fa0/11
80	Staff	active	Fa0/5
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdinet-default	active	
1005	trnet-default	active	

The CLI prompt "Switch>" is visible at the bottom of the window.

Screenshot of Router Interface Configuration (show ip interface brief):

Router0

Physical

Config

CLI

Attributes

IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.60, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.70, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.80, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

WARNING: Unauthorized access to this device is prohibited

User Access Verification

Password:

Router>enable

Password:

Router#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	unassigned	YES	manual	up	up
GigabitEthernet0/0/0.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0/0.20	192.168.10.33	YES	manual	up	up
GigabitEthernet0/0/0.30	192.168.10.65	YES	manual	up	up
GigabitEthernet0/0/0.40	192.168.10.97	YES	manual	up	up
GigabitEthernet0/0/0.50	192.168.10.129	YES	manual	up	up
GigabitEthernet0/0/0.60	192.168.10.193	YES	manual	up	up
GigabitEthernet0/0/0.70	192.168.10.161	YES	manual	up	up
GigabitEthernet0/0/0.80	192.168.10.225	YES	manual	up	up
GigabitEthernet0/0/1	8.8.8.1	YES	manual	up	up
GigabitEthernet0/0/2	unassigned	YES	manual	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

Router#

Copy

Paste

Top

Security Design

VLAN Segmentation:

VLAN segmentation was implemented for department traffic and to prevent unauthorized access between departments.

Each department has its own VLAN, ensuring that broadcast traffic is minimized and internal department communications are secure.

This configuration has improved network performance and security, ensuring that direct communication between unrelated departments is not allowed.

Access Control Lists (ACLs):

Access Control Lists were configured on the router to enforce security policies based on business requirements.

HR Access Restriction

The HR department is prevented from accessing the Finance network.

ACL Name: HR_BLOCK

Policy implemented:

- Deny HR network access to Finance network
- Allow other traffic

Guest Network Restriction

Guest users are allowed internet access only and cannot access internal company networks.

ACL Name: GUEST_ONLY_INTERNET

Policy:

- Deny access to internal networks
- Allow other traffic

Firewall Security:

Router firewall rules were implemented using ACLs to restrict inbound traffic from external networks.

Only essential services are allowed:

- HTTP (port 80)
- HTTPS (port 443)
- DNS (port 53)
- ICMP (for testing connectivity)

All other traffic is denied.

WAN_FIREWALL

Router Security:

Basic router security practices were applied including:

- Encrypted enable password
- Console password protection
- VTY password for remote access
- Message of the day banner warning unauthorized users

These measures help protect the network infrastructure from unauthorized administrative access.

Router Passwords:

First user access verification pass: cisco123

Second password after enable: class123



```
WARNING: Unauthorized access to this device is prohibited

User Access Verification

Password:

Router>enable
Password:
```

Please Note: the screenshots for VLAN segmentation, ACLs, Firewall and Router security parts are included in the Implementation and testing section.

Wireless Security:

Wireless networks are prone to security attacks, and wireless security has become a major concern for organizations using wireless networking. To prevent such security attacks, several security protocols were implemented for the wireless network configuration.

WPA2-PSK Authentication:

WPA2-PSK is the wireless security standard implemented for the wireless network configuration. This is one of the most commonly used wireless security standards, and it provides robust security for wireless networks.

WPA2 uses Advanced Encryption Standard (AES) encryption algorithm, which ensures that the wireless communication between the device and the access point is kept confidential and secure against unauthorized interception. Clients must provide the correct passphrase before they are allowed to connect to the wireless network.

Staff Vlan : 80

For the staff wireless network, the following configuration is applied:

SSID: StaffWiFi

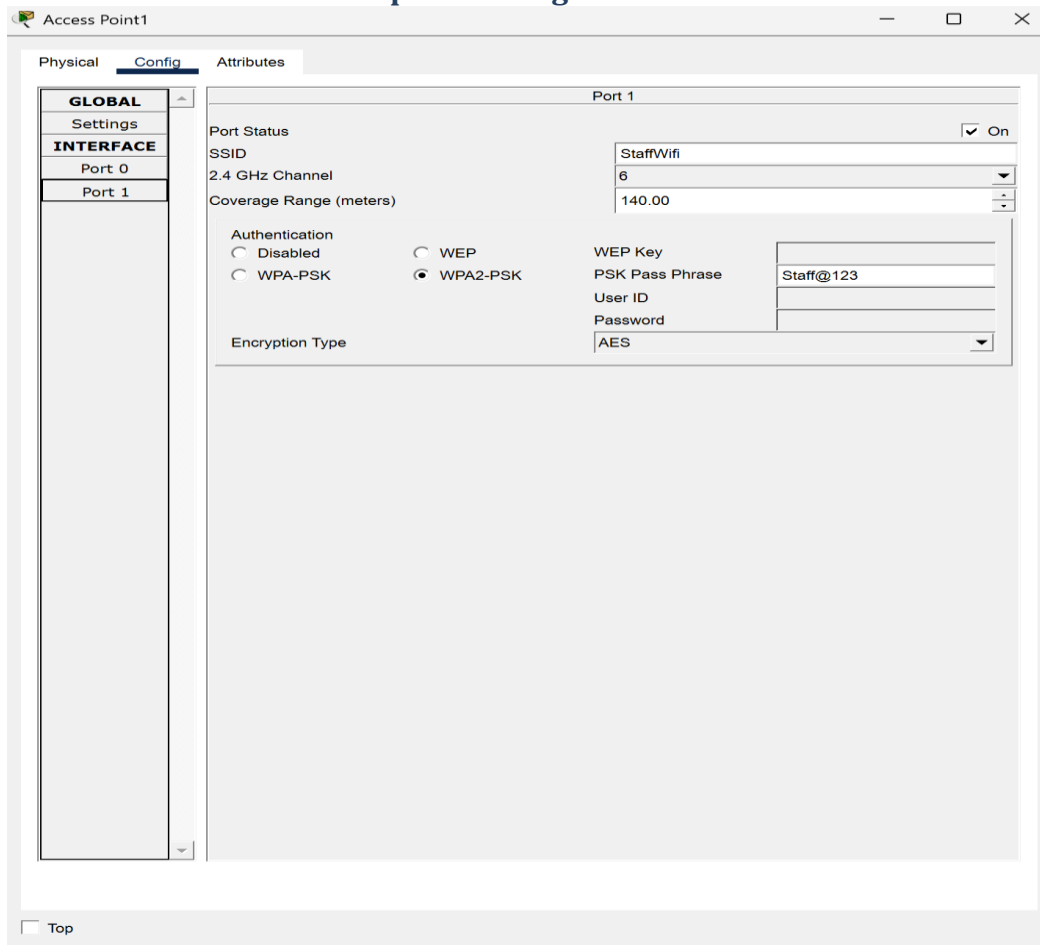
Authentication: WPA2-PSK

Encryption: AES

Passphrase: Staff@123

Only authorized staff members who are aware of the passphrase are allowed access to the staff wireless network.

Screenshot of Staff Access point Configuration:



The screenshot shows a web-based configuration interface for an access point. The window title is "Access Point1". The interface has three tabs: "Physical", "Config" (selected), and "Attributes". On the left, there is a sidebar with a tree view containing "GLOBAL" and "INTERFACE". Under "INTERFACE", "Port 0" and "Port 1" are listed. The main content area is titled "Port 1" and contains the following configuration options:

- Port Status: ☒ On
- SSID: StaffWifi
- 2.4 GHz Channel: 6
- Coverage Range (meters): 140.00
- Authentication:
 - ☐ Disabled
 - ☐ WEP
 - ☒ WPA2-PSK
- WEP Key: (empty field)
- PSK Pass Phrase: Staff@123
- User ID: (empty field)
- Password: (empty field)
- Encryption Type: AES

At the bottom left, there is a "Top" link.

Guest Wireless Isolation:

For the guest, a separate wireless network is created with a different SSID. This wireless network is isolated from the internal corporate network by using VLANs.

For the guest network, VLAN 70 is assigned. This configuration only allows the guest network access to the internet.

Screenshot of Guest Access point configuration:

The screenshot shows a web-based configuration interface for an 'Access Point2'. The window has three tabs: 'Physical', 'Config' (selected), and 'Attributes'. On the left, a sidebar contains a tree view with 'GLOBAL' (expanded) and 'INTERFACE' (containing 'Port 0' and 'Port 1'). The main area displays the configuration for 'Port 1'. The 'Port Status' is 'On'. The 'SSID' is 'GuestWifi'. The '2.4 GHz Channel' is '6'. The 'Coverage Range (meters)' is '140.00'. Under 'Authentication', 'WPA2-PSK' is selected. The 'WEP Key' field is empty. The 'PSK Pass Phrase' is 'Guest@123'. The 'User ID' and 'Password' fields are empty. The 'Encryption Type' is 'AES'. A 'Top' link is at the bottom left.

Port 1	
Port Status	☒ On
SSID	GuestWifi
2.4 GHz Channel	6
Coverage Range (meters)	140.00
Authentication	
☐ Disabled	☐ WEP
☐ WPA-PSK	☒ WPA2-PSK
WEP Key	
PSK Pass Phrase	Guest@123
User ID	
Password	
Encryption Type	AES

Access Point Security Configuration:

The access points were configured to broadcast specific SSIDs for different user groups. Security configurations were applied to allow only authenticated users to access the wireless network.

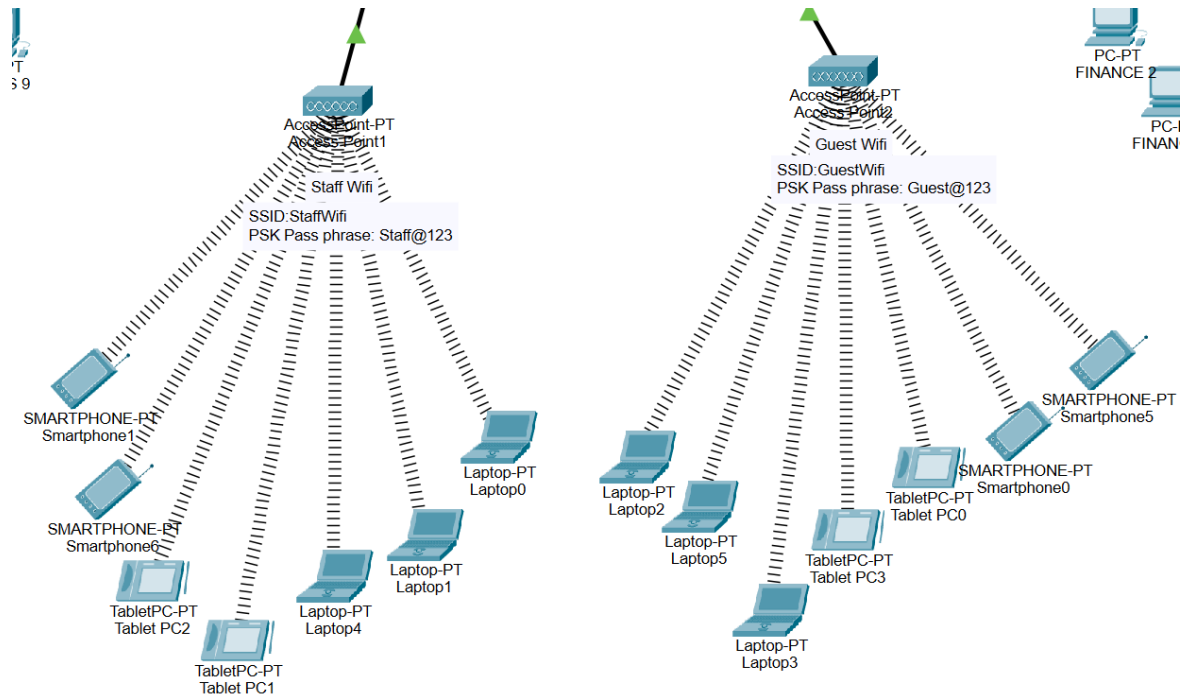
The key security configurations applied to the access points are as follows:

- WPA2-PSK authentication is enabled
- AES encryption is used for secure communication
- Different SSIDs for staff and guest users
- VLAN segmentation is applied for guest users

Protection Against Unauthorized Access:

By configuring WPA2 encryption, the wireless network is secured against different wireless security threats, including:

- Unauthorized access
- Packet sniffing
- Data interception
- Internal network exposure for guest users



Implementation & Testing

Router-on-a-Stick Implementation:

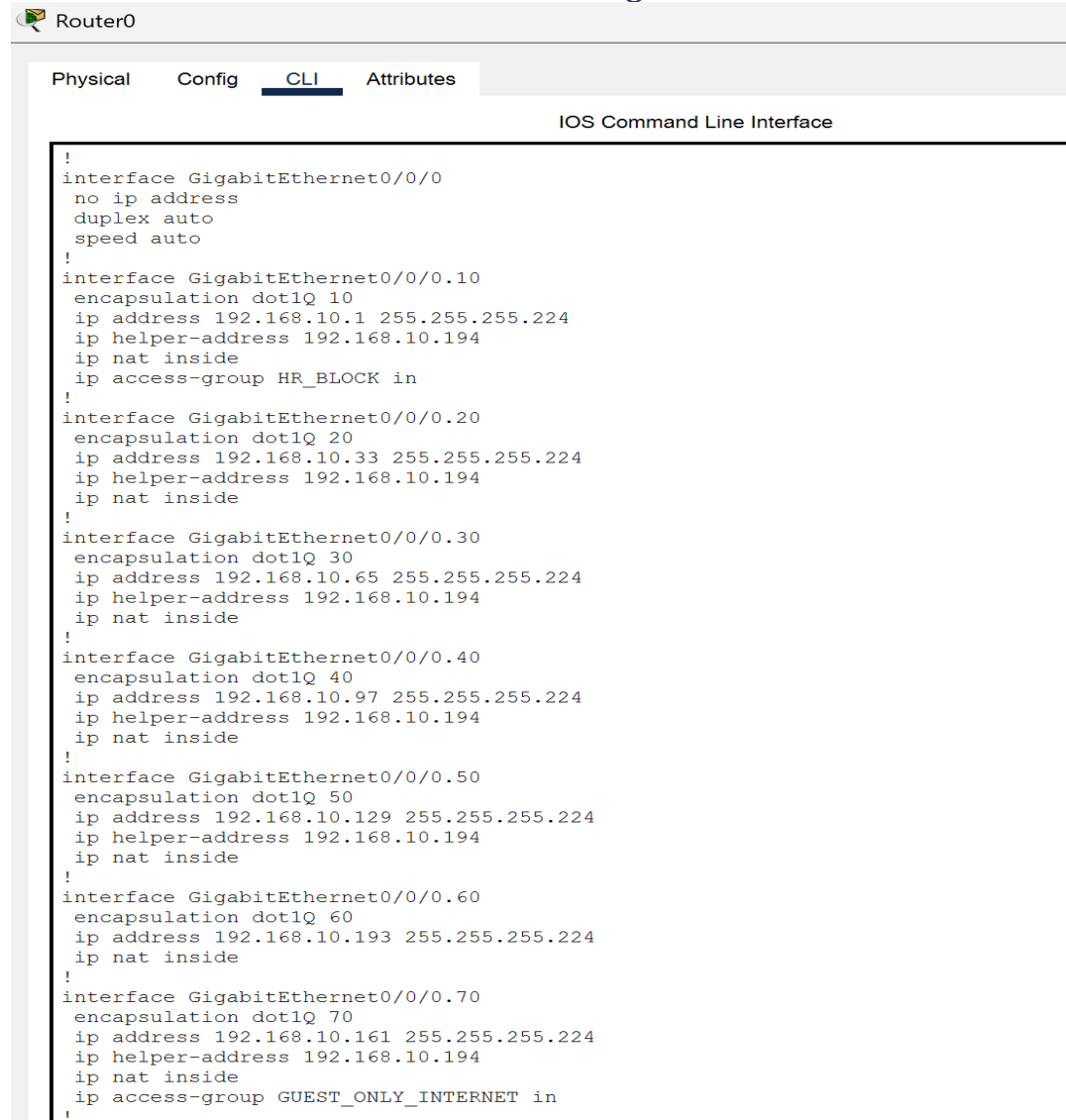
Inter-VLAN routing was achieved using the Router on a Stick configuration, where the physical interface of the router was configured for routing between multiple VLANs.

To do this:

- Sub interfaces were configured on the router's interface (G0/0/0.10, G0/0/0.20, G0/0/0.30, G0/0/0.40, G0/0/0.50).
- Each sub interface was configured for 802.1q encapsulation (dot1q) so that the router can understand the VLAN tags from the switches.
- Each VLAN was configured for a different IP address for the gateway for each department (HR, Finance, IT, Sales, and Management).

This configuration allows the router to act as the default gateway for all VLANs and allows it to route between all the departments.

Screenshots of Router Sub interfaces Configuration:



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

!
interface GigabitEthernet0/0/0
  no ip address
  duplex auto
  speed auto
!
interface GigabitEthernet0/0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.224
  ip helper-address 192.168.10.194
  ip nat inside
  ip access-group HR_BLOCK in
!
interface GigabitEthernet0/0/0.20
  encapsulation dot1Q 20
  ip address 192.168.10.33 255.255.255.224
  ip helper-address 192.168.10.194
  ip nat inside
!
interface GigabitEthernet0/0/0.30
  encapsulation dot1Q 30
  ip address 192.168.10.65 255.255.255.224
  ip helper-address 192.168.10.194
  ip nat inside
!
interface GigabitEthernet0/0/0.40
  encapsulation dot1Q 40
  ip address 192.168.10.97 255.255.255.224
  ip helper-address 192.168.10.194
  ip nat inside
!
interface GigabitEthernet0/0/0.50
  encapsulation dot1Q 50
  ip address 192.168.10.129 255.255.255.224
  ip helper-address 192.168.10.194
  ip nat inside
!
interface GigabitEthernet0/0/0.60
  encapsulation dot1Q 60
  ip address 192.168.10.193 255.255.255.224
  ip nat inside
!
interface GigabitEthernet0/0/0.70
  encapsulation dot1Q 70
  ip address 192.168.10.161 255.255.255.224
  ip helper-address 192.168.10.194
  ip nat inside
  ip access-group GUEST_ONLY_INTERNET in
!
```

```
interface GigabitEthernet0/0/0.80
 encapsulation dot1Q 80
 ip address 192.168.10.225 255.255.255.224
 ip helper-address 192.168.10.194
!
interface GigabitEthernet0/0/1
 ip address 8.8.8.1 255.255.255.0
 ip nat outside
 ip access-group WAN_FIREWALL in
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/2
 description UNUSED_PORT_SECURED
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
```

VLAN and Trunking Configuration:

To allow the VLAN traffic to pass between the switches and the router, trunk cables are configured between the switches.

To do this:

- VLAN Configuration: VLAN10, VLAN20, VLAN30, VLAN40, and VLAN50 were configured on all switches for the different departments.
- Trunk Configuration: switchport mode trunk was configured between the switches for the connections between the switches, allowing multiple VLANs to be transported over one cable.
- Encapsulation Configuration: switchport trunk encapsulation dot1q was configured on the multilayer switch (Cisco3560) before setting the trunk mode, since it was rejected without it.

Access Ports and Physical Topology:

The end devices were connected to the network through the access ports, which were configured based on the VLAN.

The steps to configure the network were as follows:

- Access Port Assignment:**

The ports for the PCs, for example, Fa0/1 to Fa0/10, were configured to the respective VLAN based on the departments by using the following commands:

```
switchport mode access
```

```
switchport access vlan VLAN ID
```

- Physical Uplink Separation:**

The uplinks were moved to higher-numbered ports such as Fa0/24 and Gigabit ports to separate the backbone from the end users.

- Default Gateway Configuration:**

The PCs were configured with the correct default gateway, which points to the router's sub interface.

NAT Configuration:

Network Address Translation (NAT) was implemented to allow internal private network devices to access the simulated internet.

Verification was performed using:

```
show ip nat translations
```

Screenshot - NAT Translation Table:

```
WARNING: Unauthorized access to this device is prohibited

User Access Verification

Password:

Router>enable
Password:
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 8.8.8.1:1024      192.168.10.164:4  8.8.8.8:4          8.8.8.8:1024
icmp 8.8.8.1:1025      192.168.10.164:5  8.8.8.8:5          8.8.8.8:1025
icmp 8.8.8.1:1026      192.168.10.164:6  8.8.8.8:6          8.8.8.8:1026
icmp 8.8.8.1:3         192.168.10.11:3   8.8.8.8:3          8.8.8.8:3
icmp 8.8.8.1:4         192.168.10.11:4   8.8.8.8:4          8.8.8.8:4
icmp 8.8.8.1:5         192.168.10.11:5   8.8.8.8:5          8.8.8.8:5
icmp 8.8.8.1:6         192.168.10.11:6   8.8.8.8:6          8.8.8.8:6
icmp 8.8.8.1:7         192.168.10.164:7  8.8.8.8:7          8.8.8.8:7

Router#
```

Network Design

The design consists of three layers:

Core/Distribution Layer:

A Cisco 3560 Multilayer Switch (Main Switch) acts as the central point connecting all departmental switches and routing traffic toward the router.

Access Layer:

Five switches were deployed for the following departments:

- HR
- Finance
- IT
- Sales
- Management

These switches provided network connectivity to end-user PCs.

Routing Layer:

A Cisco ISR 4331 Router provides:

- Inter-VLAN routing
- NAT services
- Connectivity to the server network and simulated internet

Connectivity Testing

Several tests were conducted to verify correct network functionality.

Successful tests included:

Gateway Connectivity:

Each PC successfully pinged its VLAN gateway on the router, confirming correct IP configuration.

Inter-VLAN Communication:

Devices from different departments successfully communicated through the router.

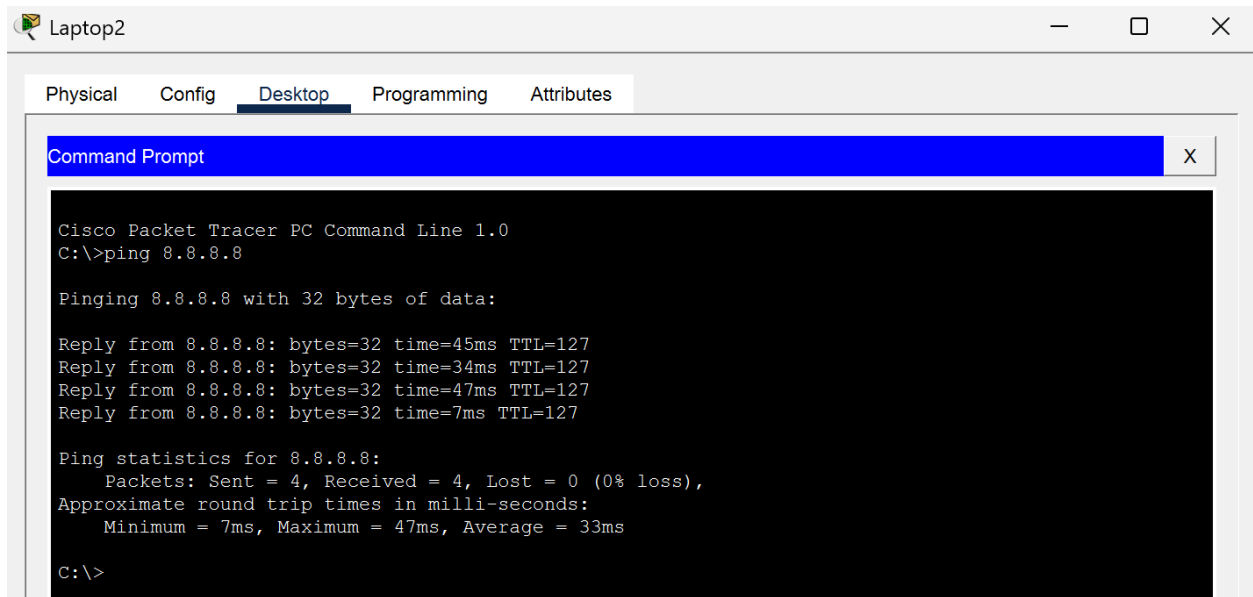
Internet Connectivity:

Internal PCs and the wireless devices successfully accessed the simulated internet server.

- PCs successfully pinged 8.8.8.8, confirming NAT and routing functionality.

Screenshots– Successful Pings to Internet (8.8.8.8):

Guest wireless device:



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

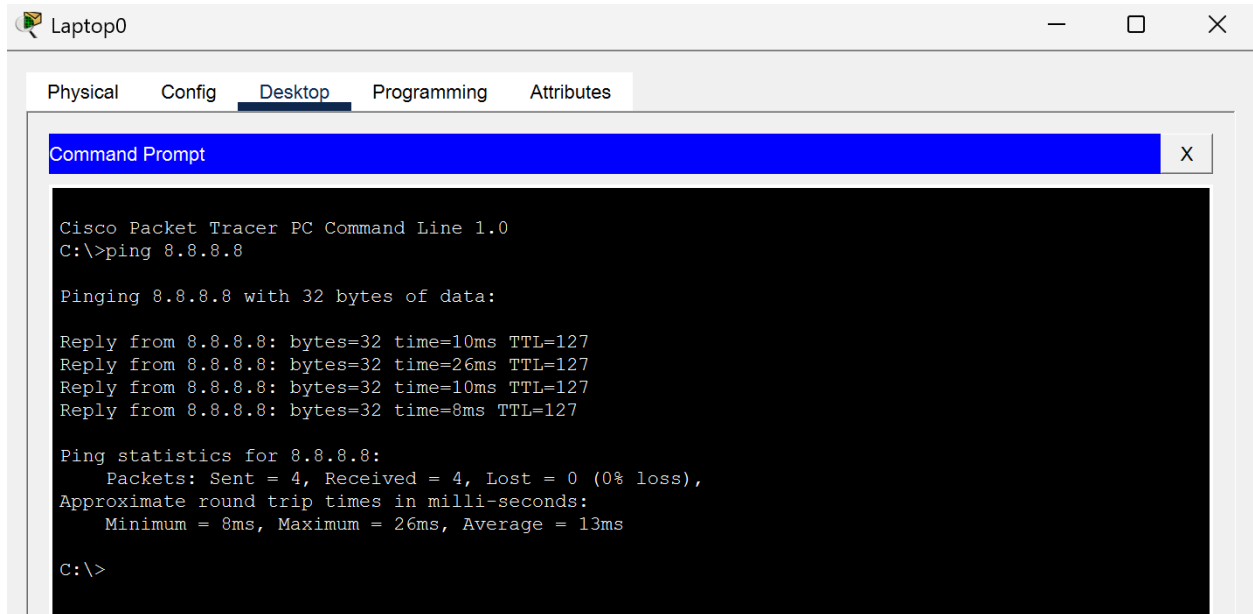
Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=45ms TTL=127
Reply from 8.8.8.8: bytes=32 time=34ms TTL=127
Reply from 8.8.8.8: bytes=32 time=47ms TTL=127
Reply from 8.8.8.8: bytes=32 time=7ms TTL=127

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 47ms, Average = 33ms

C:\>
```

Staff wireless device:



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

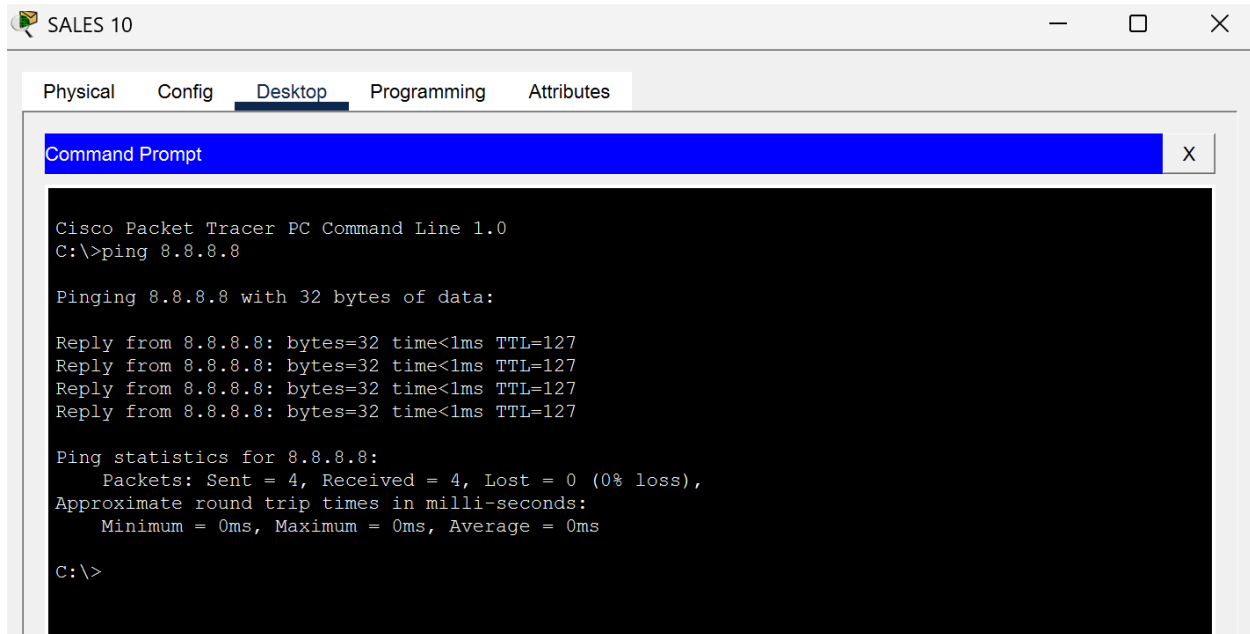
Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=10ms TTL=127
Reply from 8.8.8.8: bytes=32 time=26ms TTL=127
Reply from 8.8.8.8: bytes=32 time=10ms TTL=127
Reply from 8.8.8.8: bytes=32 time=8ms TTL=127

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 26ms, Average = 13ms

C:\>
```

Sales PC:



SALES 10

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=127
Reply from 8.8.8.8: bytes=32 time<1ms TTL=127
Reply from 8.8.8.8: bytes=32 time<1ms TTL=127
Reply from 8.8.8.8: bytes=32 time<1ms TTL=127

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Security Testing

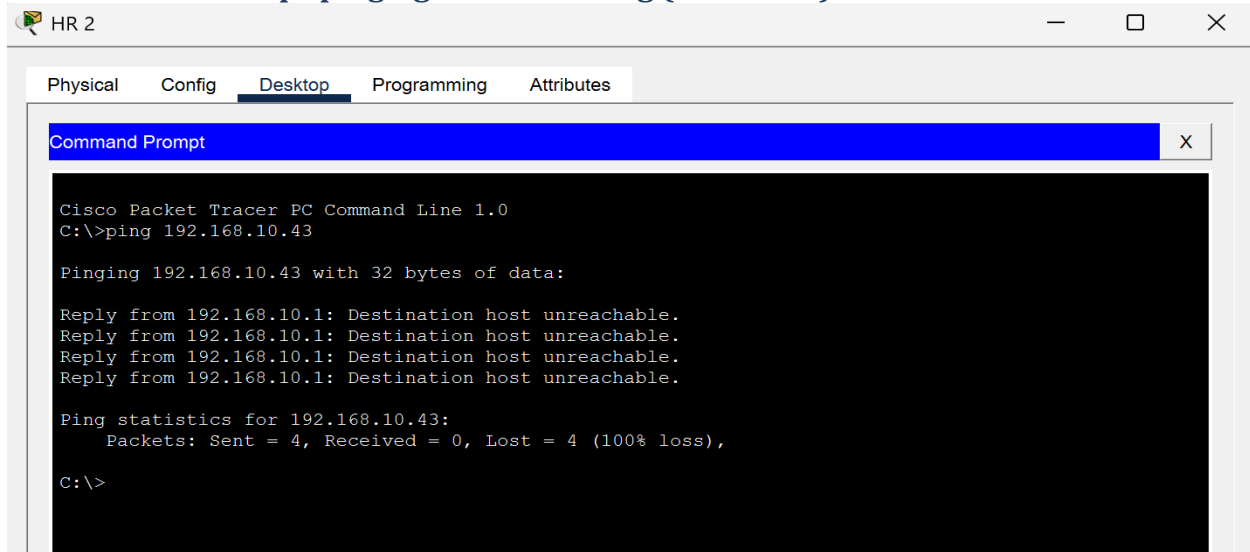
Security configurations were also verified using access control policies.

Tests confirmed that:

- HR VLAN cannot access Finance VLAN
- Guest VLAN cannot access internal corporate networks
- Only authorized traffic passes through configured ACL and firewall rules

A failed ping test confirmed that restricted traffic was successfully blocked by the security rules.

Screenshots of HR pc pinging Finance failing (ACL block):



HR 2

Physical Config **Desktop** Programming Attributes

Command Prompt

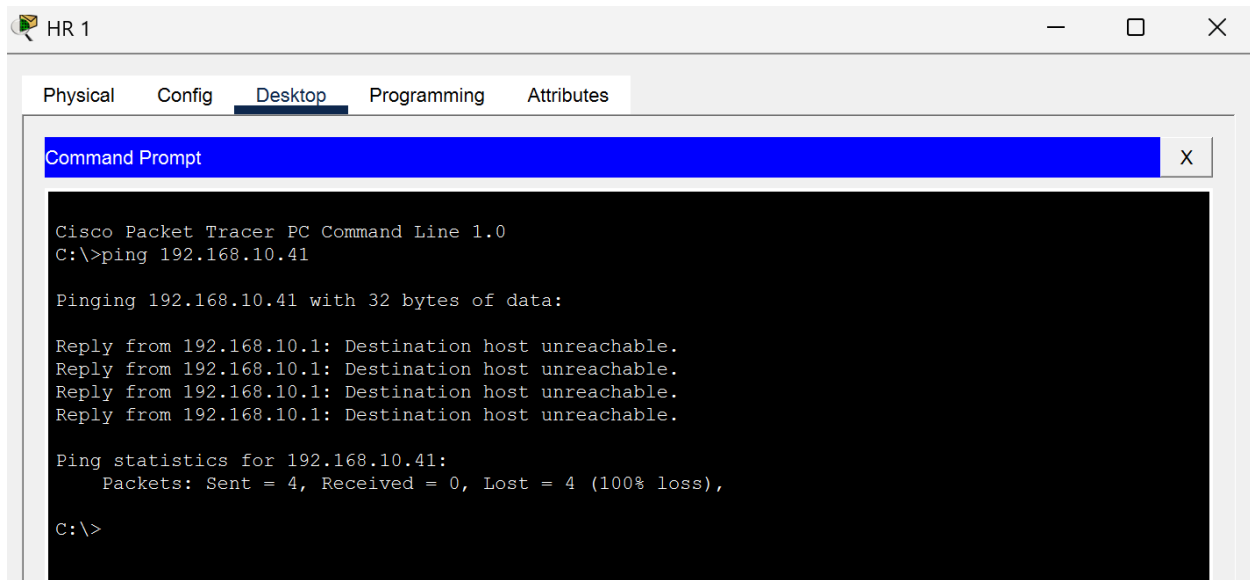
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.43

Pinging 192.168.10.43 with 32 bytes of data:

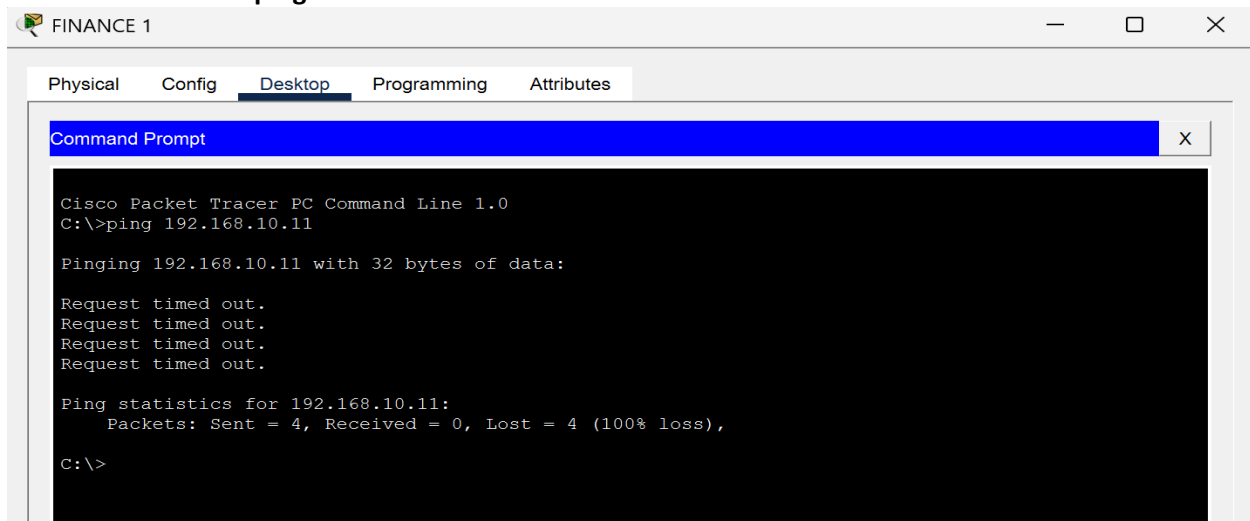
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.10.43:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

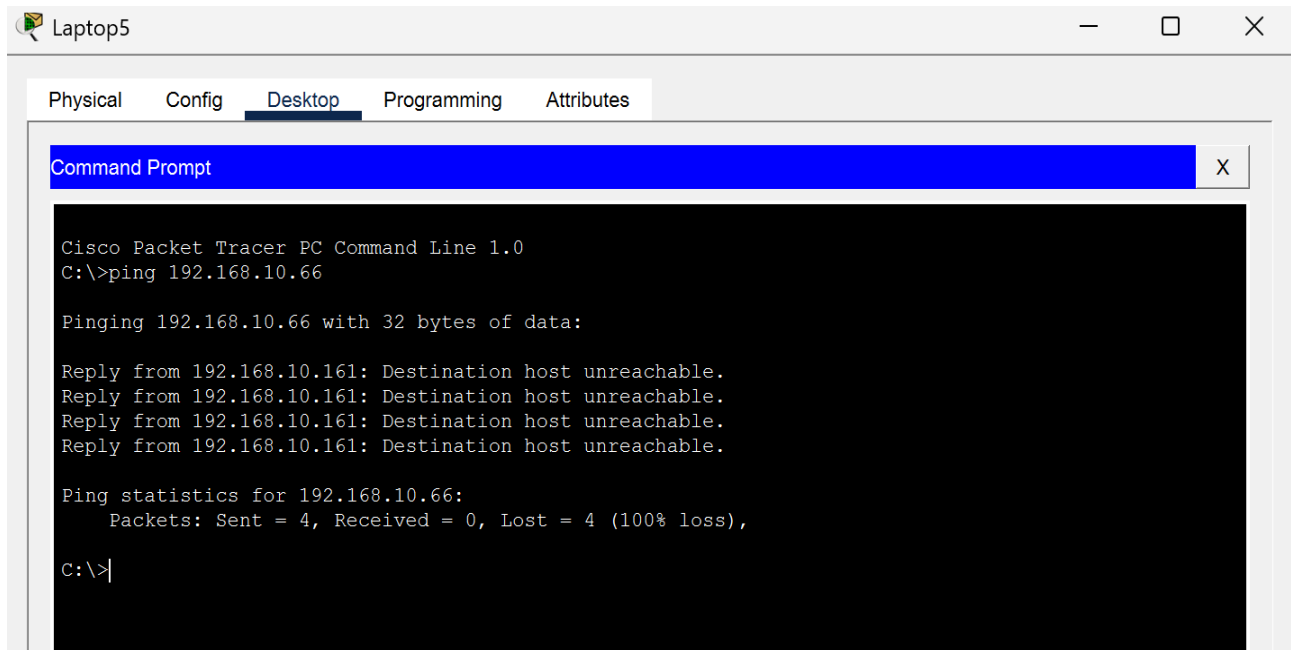
C:\>
```



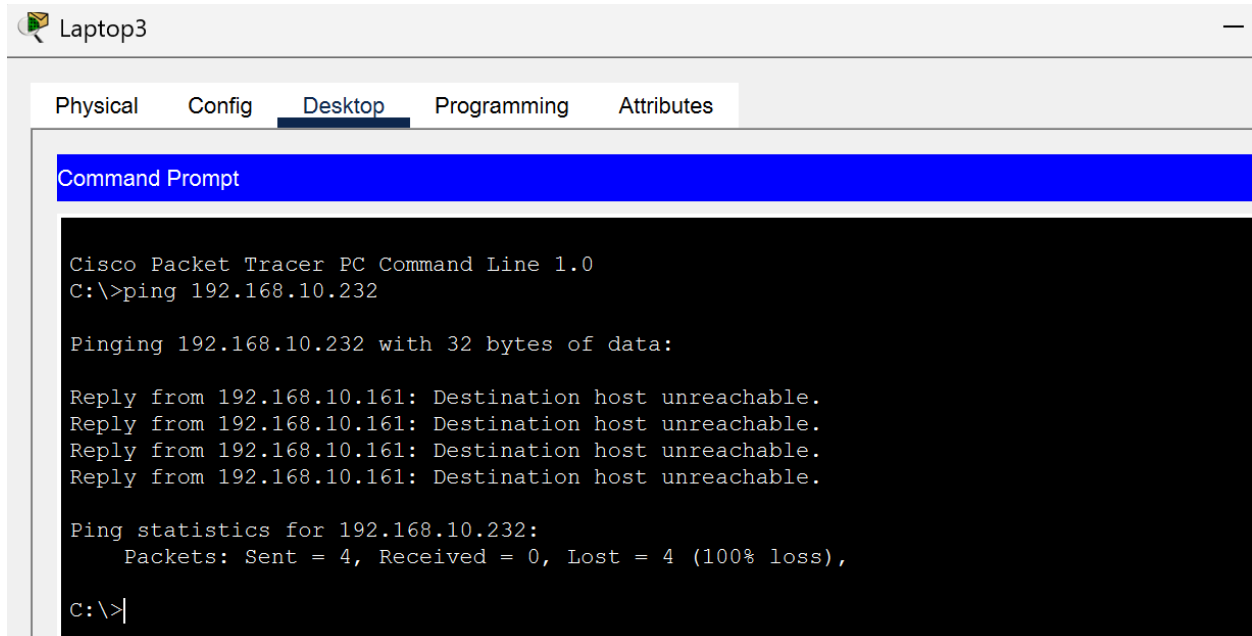
Finance also cannot ping HR:



Screenshot- Guest devices cannot ping internal devices(ACL BLOCK):



Screenshots – Guest wireless devices cannot ping Staff wireless devices(& vice versa):



 Tablet PC1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.162

Pinging 192.168.10.162 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.162:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Screenshot – Guest to Guest Successful ping:

 Laptop5

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.162

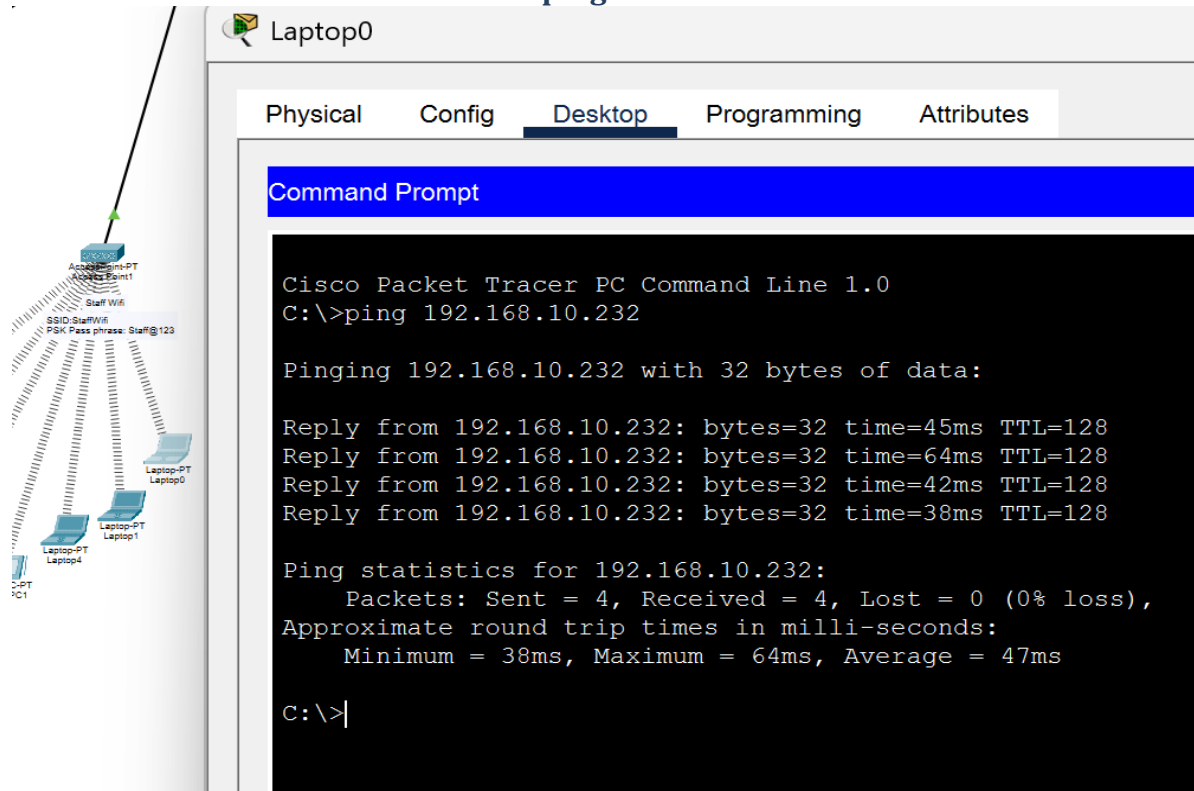
Pinging 192.168.10.162 with 32 bytes of data:

Reply from 192.168.10.162: bytes=32 time=115ms TTL=128
Reply from 192.168.10.162: bytes=32 time=8ms TTL=128
Reply from 192.168.10.162: bytes=32 time=29ms TTL=128
Reply from 192.168.10.162: bytes=32 time=67ms TTL=128

Ping statistics for 192.168.10.162:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 115ms, Average = 54ms

C:\>
```

Screenshot - Staff to Staff Successful ping:



Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.232

Pinging 192.168.10.232 with 32 bytes of data:

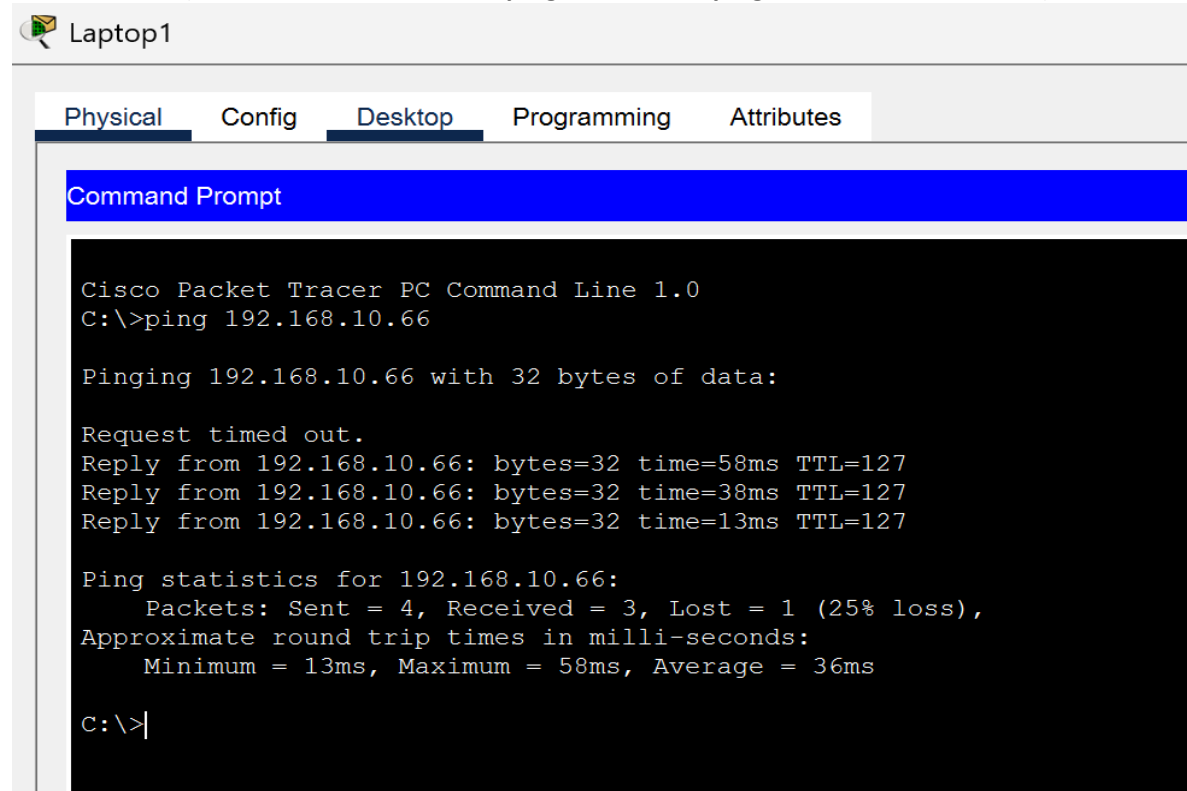
Reply from 192.168.10.232: bytes=32 time=45ms TTL=128
Reply from 192.168.10.232: bytes=32 time=64ms TTL=128
Reply from 192.168.10.232: bytes=32 time=42ms TTL=128
Reply from 192.168.10.232: bytes=32 time=38ms TTL=128

Ping statistics for 192.168.10.232:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 38ms, Maximum = 64ms, Average = 47ms

C:\>
```

Screenshot Staff wireless device to internal devices Successful ping:

Staff to IT PC: (staff wireless devices can ping and receive pings from internal devices)



Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.66

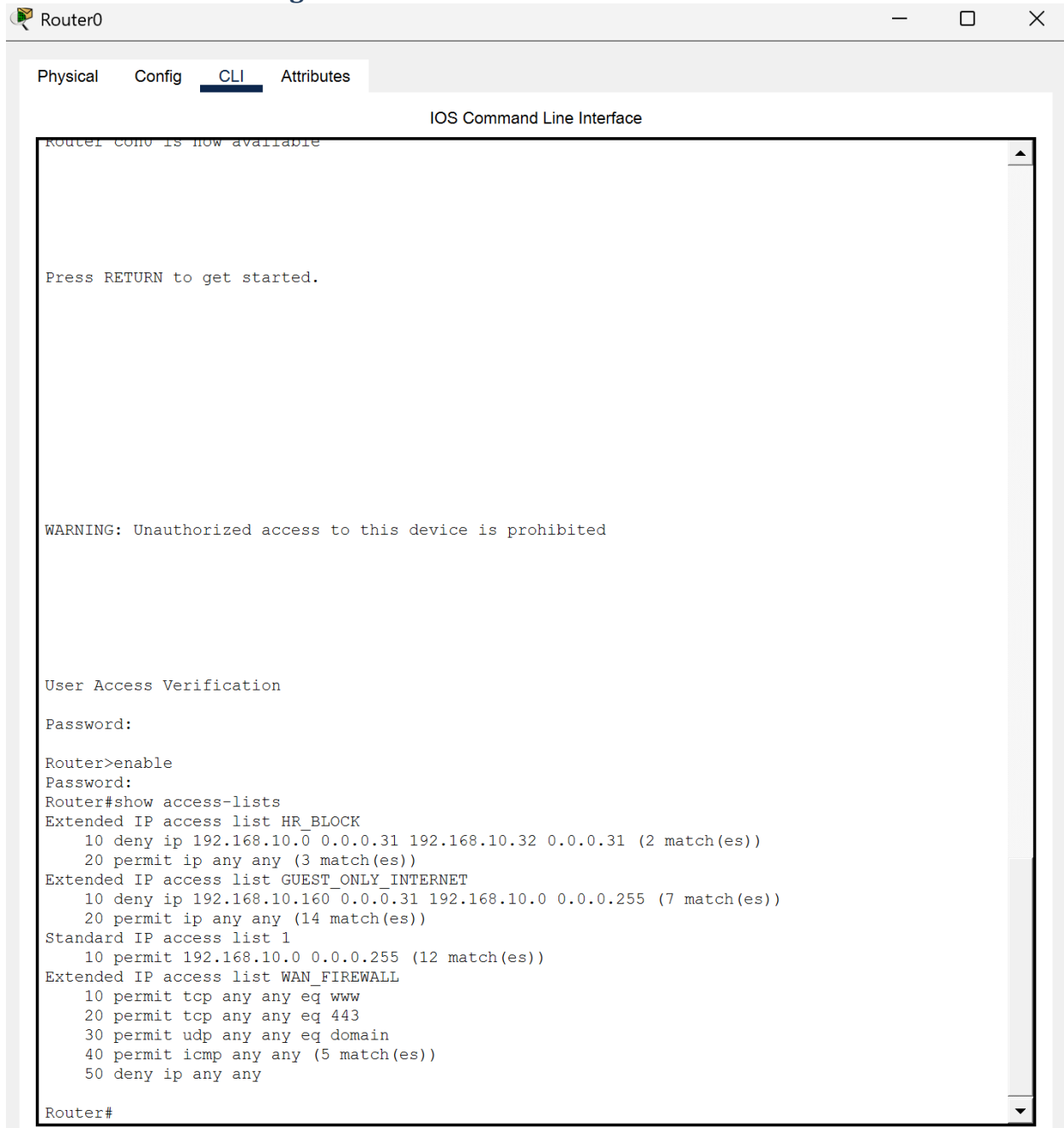
Pinging 192.168.10.66 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.66: bytes=32 time=58ms TTL=127
Reply from 192.168.10.66: bytes=32 time=38ms TTL=127
Reply from 192.168.10.66: bytes=32 time=13ms TTL=127

Ping statistics for 192.168.10.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 58ms, Average = 36ms

C:\>
```

Screenshot – ACL Configuration



Successful pings:

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	IT 1	HR 10	ICMP		0.000	N	0	(edit)	
	Successful	IT 1	SALES 1	ICMP		0.000	N	1	(edit)	
	Successful	IT 1	MANAGEMENT 10	ICMP		0.000	N	2	(edit)	
	Successful	IT 1	MANAGEMENT 10	ICMP		0.000	N	3	(edit)	
	Successful	IT 1	FINANCE 1	ICMP		0.000	N	4	(edit)	
	Successful	MANAG...	HR 10	ICMP		0.000	N	5	(edit)	
	Successful	SALES 10	MANAGEMENT 10	ICMP		0.000	N	6	(edit)	
	Successful	FINANC...	SALES 10	ICMP		0.000	N	7	(edit)	
	Successful	FINANC...	MANAGEMENT 1	ICMP		0.000	N	8	(edit)	
	Successful	HR 1	SALES 1	ICMP		0.000	N	9	(edit)	
	Successful	HR 10	MANAGEMENT 10	ICMP		0.000	N	10	(edit)	
	Successful	FINANC...	SALES 10	ICMP		0.000	N	11	(edit)	
	Successful	MANAG...	HR 10	ICMP		0.000	N	12	(edit)	
	Successful	FINANC...	IT 10	ICMP		0.000	N	13	(edit)	
	Successful	FINANC...	MANAGEMENT 10	ICMP		0.000	N	14	(edit)	
	Successful	SALES 10	IT 10	ICMP		0.000	N	15	(edit)	
	Successful	HR 6	IT 1	ICMP		0.000	N	16	(edit)	
	Successful	HR 1	internetServer	ICMP		0.000	N	17	(edit)	
	Successful	SALES 10	internetServer	ICMP		0.000	N	18	(edit)	
	Successful	IT 1	internetServer	ICMP		0.000	N	19	(edit)	
	Successful	MANAG...	internetServer	ICMP		0.000	N	20	(edit)	
	Successful	FINANC...	internetServer	ICMP		0.000	N	21	(edit)	
	Successful	Laptop2	Laptop5	ICMP		0.000	N	22	(edit)	
	Successful	Laptop3	Laptop5	ICMP		0.000	N	23	(edit)	
	Successful	Laptop3	Tablet PC3	ICMP		0.000	N	24	(edit)	
	Successful	Smartph...	Tablet PC0	ICMP		0.000	N	25	(edit)	
	Successful	Tablet PC3	Tablet PC0	ICMP		0.000	N	26	(edit)	
	Successful	Smartph...	Laptop5	ICMP		0.000	N	27	(edit)	
	Successful	Laptop2	Smartphone5	ICMP		0.000	N	28	(edit)	
	Successful	Tablet PC3	Smartphone5	ICMP		0.000	N	29	(edit)	
	Successful	Smartph...	Smartphone5	ICMP		0.000	N	30	(edit)	
	Successful	Laptop0	Laptop1	ICMP		0.000	N	31	(edit)	
	Successful	Laptop4	Laptop1	ICMP		0.000	N	32	(edit)	
	Successful	Laptop1	Tablet PC1	ICMP		0.000	N	33	(edit)	
	Successful	Tablet PC2	Laptop1	ICMP		0.000	N	34	(edit)	
	Successful	Tablet PC1	Tablet PC2	ICMP		0.000	N	35	(edit)	
	Successful	Tablet PC1	Smartphone6	ICMP		0.000	N	36	(edit)	
	Successful	Smartph...	Smartphone6	ICMP		0.000	N	37	(edit)	
	Successful	Smartph...	Tablet PC2	ICMP		0.000	N	38	(edit)	
	Successful	SALES 7	companyServer	ICMP		0.000	N	62	(edit)	
	Successful	HR 1	companyServer	ICMP		0.000	N	63	(edit)	
	Successful	FINANC...	companyServer	ICMP		0.000	N	64	(edit)	
	Successful	MANAG...	companyServer	ICMP		0.000	N	65	(edit)	
	Successful	IT 10	companyServer	ICMP		0.000	N	66	(edit)	
	Successful	Router0	company...	IC...		0.000	N	6	(e...	(delete)
	Successful	Router0	internetSe...	IC...		0.000	N	7	(e...	(delete)

	Successful	Laptop4	MANAGEMENT 10	ICMP		0.000	N	40	(edit)
	Successful	Tablet PC2	HR 1	ICMP		0.000	N	41	(edit)
	Successful	Smartph...	MANAGEMENT 1	ICMP		0.000	N	42	(edit)
	Successful	Laptop1	IT 10	ICMP		0.000	N	43	(edit)
	Successful	IT 8	Laptop4	ICMP		0.000	N	44	(edit)
	Successful	MANAG...	Laptop1	ICMP		0.000	N	45	(edit)
	Successful	FINANC...	Laptop1	ICMP		0.000	N	46	(edit)
	Successful	Laptop0	internetServer	ICMP		0.000	N	47	(edit)
	Successful	Laptop4	internetServer	ICMP		0.000	N	48	(edit)
	Successful	Tablet PC1	internetServer	ICMP		0.000	N	49	(edit)
	Successful	Smartph...	internetServer	ICMP		0.000	N	50	(edit)
	Successful	Laptop2	internetServer	ICMP		0.000	N	51	(edit)
	Successful	Tablet PC3	internetServer	ICMP		0.000	N	52	(edit)
	Successful	Smartph...	internetServer	ICMP		0.000	N	53	(edit)
	Successful	Tablet PC1	SALES 10	ICMP		0.000	N	54	(edit)
	Successful	Laptop0	IT 10	ICMP		0.000	N	55	(edit)
	Successful	Laptop1	HR 1	ICMP		0.000	N	56	(edit)
	Successful	Smartph...	FINANCE 2	ICMP		0.000	N	57	(edit)
	Successful	Smartph...	MANAGEMENT 10	ICMP		0.000	N	58	(edit)
	Successful	Laptop4	HR 10	ICMP		0.000	N	59	(edit)
	Successful	Tablet PC1	FINANCE 2	ICMP		0.000	N	60	(edit)
	Successful	Smartph...	IT 9	ICMP		0.000	N	61	(edit)

Failed pings-ACL Blocking Test:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	HR 1	FINANCE 1	IC...		0.000	N	0	(e...	(delete)
	Failed	HR 10	FINANCE 7	IC...		0.000	N	1	(e...	(delete)
	Failed	FINA...	HR 7	IC...		0.000	N	2	(e...	(delete)
	Failed	FINA...	HR 5	IC...		0.000	N	3	(e...	(delete)

	Failed	Laptop2	companyServer	ICMP		0.000	N	67	(edit)
	Failed	Tablet PC3	companyServer	ICMP		0.000	N	68	(edit)
	Failed	Laptop2	SALES 8	ICMP		0.000	N	69	(edit)
	Failed	Laptop3	HR 1	ICMP		0.000	N	70	(edit)
	Failed	Laptop5	MANAGEMENT 1	ICMP		0.000	N	71	(edit)
	Failed	Smartph...	HR 10	ICMP		0.000	N	72	(edit)
	Failed	Smartph...	IT 9	ICMP		0.000	N	73	(edit)
	Failed	Laptop2	Laptop0	ICMP		0.000	N	74	(edit)
	Failed	Laptop5	Tablet PC1	ICMP		0.000	N	75	(edit)
	Failed	Smartph...	Smartphone1	ICMP		0.000	N	76	(edit)
	Failed	Tablet PC0	Tablet PC1	ICMP		0.000	N	77	(edit)
	Failed	Tablet PC3	Tablet PC2	ICMP		0.000	N	78	(edit)
	Failed	MANAG...	Laptop5	ICMP		0.000	N	79	(edit)
	Failed	IT 10	Laptop2	ICMP		0.000	N	80	(edit)
	Failed	HR 10	Smartphone0	ICMP		0.000	N	81	(edit)
	Failed	FINANC...	Laptop2	ICMP		0.000	N	82	(edit)
	Failed	SALES 1	Smartphone5	ICMP		0.000	N	83	(edit)

- HR and Finance cannot ping each other.
- Guest wireless devices cannot ping internal company devices nor can they receive ping from internal company devices.
- Guest and staff wireless devices cannot ping each other.

Screenshot of Server DHCP:

companyServer

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

Interface

FastEthernet0

Service

On

Off

Pool Name

serverPool

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

Start IP Address :

192

168

10

192

Subnet Mask:

255

255

255

224

Maximum Number of Users :

512

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Staff	192.168.10.225	192.168.10.194	192.168.10.226	255.255.255.224	25	0.0.0.0	0.0.0.0
Guest	192.168.10.161	192.168.10.194	192.168.10.162	255.255.255.224	25	0.0.0.0	0.0.0.0
Management	192.168.10.129	192.168.10.194	192.168.10.130	255.255.255.224	25	0.0.0.0	0.0.0.0
Sales	192.168.10.97	192.168.10.194	192.168.10.98	255.255.255.224	25	0.0.0.0	0.0.0.0
IT	192.168.10.65	192.168.10.194	192.168.10.66	255.255.255.224	25	0.0.0.0	0.0.0.0
Finance	192.168.10.33	192.168.10.194	192.168.10.34	255.255.255.224	25	0.0.0.0	0.0.0.0
HR	192.168.10.1	192.168.10.194	192.168.10.2	255.255.255.224	25	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.10.192	255.255.255.224	512	0.0.0.0	0.0.0.0

Top

Router0

Physical	Config	CLI	Attributes
----------	--------	-----	------------

IOS Command

[illegible]

```
interface GigabitEthernet0/0/1
ip address 8.8.8.1 255.255.255.0
ip nat outside
ip access-group WAN_FIREWALL in
duplex auto
speed auto
!
interface GigabitEthernet0/0/2
description UNUSED_PORT_SECURED
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip nat inside source list 1 interface GigabitEthernet0/0/1 overload
ip classless
ip route 192.168.30.0 255.255.255.0 10.0.0.2
!
ip flow-export version 9
!
!
ip access-list extended HR_BLOCK
deny ip 192.168.10.0 0.0.0.31 192.168.10.32 0.0.0.31
permit ip any any
ip access-list extended GUEST_ONLY_INTERNET
deny ip 192.168.10.160 0.0.0.31 192.168.10.0 0.0.0.255
permit ip any any
access-list 1 permit 192.168.10.0 0.0.0.255
ip access-list extended WAN_FIREWALL
permit tcp any any eq www
permit tcp any any eq 443
permit udp any any eq domain
permit icmp any any
deny ip any any
!
banner motd ^C
WARNING: Unauthorized access to this device is prohibited

^C
!
!
!
!
line con 0
password 7 0822455D0A16544541
login
!
line aux 0
!
line vty 0 4
password 7 082048430017544541
login
transport input telnet
!
!
!
end
```

Screenshots for VLAN segmentation in each department switch:

HR switch

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch>show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10 HR	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
20 Finance	active	
30 IT	active	
40 Sales	active	
50 Management	active	
100 Server	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch>
```

SALES switch

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch>show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10 HR	active	
20 Finance	active	
30 IT	active	
40 Sales	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
50 Management	active	
100 Server	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch>
```

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10	HR	active	
20	Finance	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
30	IT	active	
40	Sales	active	
50	Management	active	
100	Server	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>
```

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10	HR	active	
20	Finance	active	
30	IT	active	
40	Sales	active	
50	Management	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
100	Server	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>
```

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10	HR	active	
20	Finance	active	
30	IT	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
40	Sales	active	
50	Management	active	
100	Server	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>
```

Screenshot of the Routing Table:



Router0

Physical Config CLI Attributes

IOS Command Line Interface

WARNING: Unauthorized access to this device is prohibited

User Access Verification

Password:

Router>enable

Password:

Router#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
      8.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       8.8.8.0/24 is directly connected, GigabitEthernet0/0/1
L       8.8.8.1/32 is directly connected, GigabitEthernet0/0/1
      192.168.10.0/24 is variably subnetted, 16 subnets, 2 masks
C       192.168.10.0/27 is directly connected, GigabitEthernet0/0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0.10
C       192.168.10.32/27 is directly connected, GigabitEthernet0/0/0.20
L       192.168.10.33/32 is directly connected, GigabitEthernet0/0/0.20
C       192.168.10.64/27 is directly connected, GigabitEthernet0/0/0.30
L       192.168.10.65/32 is directly connected, GigabitEthernet0/0/0.30
C       192.168.10.96/27 is directly connected, GigabitEthernet0/0/0.40
L       192.168.10.97/32 is directly connected, GigabitEthernet0/0/0.40
C       192.168.10.128/27 is directly connected, GigabitEthernet0/0/0.50
L       192.168.10.129/32 is directly connected, GigabitEthernet0/0/0.50
C       192.168.10.160/27 is directly connected, GigabitEthernet0/0/0.70
L       192.168.10.161/32 is directly connected, GigabitEthernet0/0/0.70
C       192.168.10.192/27 is directly connected, GigabitEthernet0/0/0.60
L       192.168.10.193/32 is directly connected, GigabitEthernet0/0/0.60
C       192.168.10.224/27 is directly connected, GigabitEthernet0/0/0.80
L       192.168.10.225/32 is directly connected, GigabitEthernet0/0/0.80
```

Router#

Discussion & Evaluation

For a medium-sized IT services company, this project effectively developed and simulated an enterprise-level network that is safe, scalable, and efficient. The logical network architecture was designed to accommodate both present functional demands and those of future development. The network enables effective communication while enforcing high security measures by putting in place routers, Layer 2/Layer 3 switches, VLAN segmentation, wireless access points, and crucial servers like DHCP and DNS.

A systematic IP addressing scheme with private addressing was set up along with appropriate subnetting for each department. This method not only enhanced address use but also enhanced network control and organization. VLAN segmentation was crucial for segregating departmental traffic (HR, Finance, IT, Sales, and Management), lowering broadcast domains, and improving overall performance. Inter-VLAN routing was carefully set up to allow only business-related communication between departments.

Security was addressed as an essential requirement throughout the design phase. Access Control Lists (ACLs) were utilized to impose strict access restrictions such as banning HR from accessing Finance servers and limiting Guest VLAN users to internet access. The IT department was given full administrative access, as stated. Router and firewall security configurations were utilized to block unnecessary ports, limit incoming data traffic, and allow only essential services like HTTP, HTTPS, and DNS. WiFi security was strengthened with WPA2/WPA3 encryption standards and different SSIDs for workers and visitors to enable regulated and secure connectivity.

The topology design took network dependability and scalability into account, allowing for future additions of departments, devices, and services without requiring extensive reorganization. Simulation with Cisco Packet Tracer validated network functionality by demonstrating successful communication between authorized networks, blocked traffic where ACL policies were enforced, proper routing table configuration, internet connectivity via cloud simulation, and secure wireless device connections.

Overall, the implemented architecture adheres to business requirements by balancing performance, security, and scalability. The project exhibits a thorough knowledge of network architectural concepts, security best practices, and practical implementation strategies needed in current business systems.

Conclusion

This project required designing and modelling a structured enterprise network for a medium-sized firm. The architecture divided departments into distinct VLANs to better traffic control and internal security. A private IP addressing system with appropriate subnetting enabled easy administration and further development. Routing policies limited communication between departments and allowed only authorized access. ACLs, restricted service permissions, and separate WiFi networks for workers and visitors all contributed to increased security. Core services such as DHCP and DNS enabled seamless functioning. Cisco Packet Tracer testing demonstrated adequate connection, routing correctness, and the successful application of security rules.

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Work Distribution:

Student Name	Student ID	Role / Responsibility	Contribution Description	Percentage Contribution	Signature
Muhammad Haseeb	8874165	Group leader	DHCP configuration, router configuration, router security and firewall, wireless access point and wireless security, edited logical network design, server configuration, internet server for internet simulation, Network security part in the report, presentation video recording.	20%	MH
Rishabh Bhaskar	8984815	Member 1	VLAN segmentation, edited logical Network Design, Wireless Access Point and Wireless Security, Switch Configuration, Access Control Lists, title page, editing the report, Network Design part in report, Network Security part in the report, Implementation & testing part and all screenshots in the report.	20%	RB
Muhammad Saad Shakeel	8188430	Member 2	Created logical Network Design, IP addressing plan, subnetting, selected routing protocol, introduction and objectives, Discussion & Evaluation and conclusion.	20%	MS
Vatsh Parmar	1089067	Scribe	Business Requirement Analysis justified design decisions, conclusion, and finalized report formatting and references, VLANs for department segregation.	20%	VP
Rami Haddad	9176391	Member 3	Created logical Network Design, VLANs for department segregation, IP addressing plan, implementation and testing of initial network design getting successful pings.	20%	RH